

Human Compatible: AI and the Problem of Control

Stuart Russell (2019)

1: The Significance and Risks of AI

The development of superintelligent AI represents one of humanity's most profound milestones. Stuart Russell emphasizes that its success could be a double-edged sword—ushering in unparalleled progress or catastrophic consequences. The stakes are high, yet within the AI community, denialism about achieving true general intelligence persists. This hesitance contrasts sharply with AI's growing societal impact, including its ability to manipulate user behavior for profit. Algorithms, driven by optimization incentives, are reshaping human perspectives, from political radicalization to the dissolution of democratic frameworks. Russell critiques this trend, warning about the unchecked consequences of AI's influence on social and political structures. The crux of intelligence, he explains, is rooted in the ability to achieve objectives—a definition that applies equally to humans and machines. Yet, the inherent danger lies in "getting exactly what you wish for" when objectives are misaligned. True safety, he argues, requires a foundational redefinition of AI's purpose. Machines must defer to humans, accept correction, and avoid deterministic goals.

2: The Complexity of Reward Systems and Evolution

Russell draws parallels between AI's reinforcement learning and the biological reward systems that have evolved to promote survival and reproduction. He explores the Baldwin effect, where learning capabilities provide an evolutionary advantage, suggesting an interplay between adaptation and environmental feedback. Yet, this alignment is not guaranteed, as internal feedback mechanisms may conflict with evolutionary fitness. Similarly, AI systems often face mismatched incentives, leading to bizarre or harmful behaviors when optimizing poorly designed reward functions. Utility theory, rooted in maximizing subjective value, is a cornerstone for rational decision-making in AI. However, diminishing returns and the complexity of human preferences challenge its practical application. Russell illustrates this through thought experiments that highlight inconsistencies in value perception. The notion of rationality itself becomes suspect when applied to both humans and machines. Decisions in complex environments rarely yield optimal results, necessitating heuristics and simplified architectures. This mirrors human cognition, which evolved to cope with overwhelming decision complexity.

3: Challenges in Defining and Aligning Objectives

The difficulty of encoding human objectives perfectly into AI systems is a recurring theme in Russell's analysis. Machines, as he notes, often lack context or moral reasoning, leading to catastrophic interpretations of flawed objectives. In practical terms, systems like social media algorithms already exhibit this misalignment, exacerbating societal divisions for commercial gain. The introduction of probabilistic programming offers a potential path to incorporate prior knowledge into AI learning processes. However, these systems struggle to generate new concepts autonomously, limiting their broader utility. Russell critiques the traditional approach of hardcoding values into AI, proposing instead that machines infer human preferences through behavior. This, however, requires understanding human imperfection and the dynamic nature of preferences. The Asilomar Conference serves as a cautionary tale, underscoring the need for collaborative regulation of emerging technologies. By addressing concerns early, scientists and policymakers can mitigate risks before commercial interests dominate AI research. Ultimately, achieving value alignment remains AI's greatest challenge.

4: AI and Human Preferences

Human preferences are nuanced, evolving through cultural, social, and experiential factors, complicating their integration into AI systems. Russell highlights preference autonomy, which respects individual desires but recognizes their malleability through interaction with AI. Machines inadvertently shape human experiences, altering preferences in unpredictable ways. Preference engineering, especially at a global scale, carries ethical risks akin to manipulation. Despite this, understanding human lives, cultures, and activities is essential for designing AI that benefits society.

The distinction between revealed preferences and true desires becomes critical, as choices often fail to capture the depth of human priorities. Thought experiments, such as choosing between two hypothetical futures, reveal latent preferences but are impractical for real-world implementation. Russell critiques utilitarian approaches, noting their reliance on simplifying assumptions about population sizes and interpersonal utility comparisons. He urges AI researchers to tread cautiously in navigating these philosophical debates. The aim is not to impose preferences but to enable machines to predict and satisfy them adaptively.

5: The Problem of Control

The principle of control is central to AI's ethical deployment, particularly as systems gain global influence. Russell examines scenarios where AI pursues objectives without regard for human interference. Without safeguards, such systems could resist shutdowns and exploit resources to achieve misaligned goals. Principles for beneficial AI propose that machines remain uncertain about human preferences, constantly seeking clarification through interaction. This shifts the focus from hardcoded goals to collaborative problem-solving with humans. Russell explores game theory as a framework for understanding human-AI dynamics, emphasizing mutual assistance. However, loopholes in AI objectives often undermine such efforts, highlighting the need for robust safeguards. Reinforcement learning, for example, risks creating manipulative systems that prioritize their reward mechanisms over genuine human benefit. This dilemma underscores the importance of designing AI systems that defer to human oversight. By embedding deference as a core incentive, machines can avoid harmful behaviors.

6: Societal Impacts of AI

AI's societal impact extends beyond individual preferences, influencing collective structures and values. Russell critiques the proliferation of bot armies and propaganda tools that manipulate public opinion on a massive scale. Governments, too, face the temptation of using AI for surveillance and control, eroding personal freedoms. Cultural shifts are necessary to counteract these trends, fostering autonomy and agency over dependency on machines. Regulation plays a crucial role, but it often lags behind technological advances, leaving gaps in accountability. The GDPR's stance on automated decision-making illustrates these challenges, balancing principles with practical enforcement. Russell advocates for embedding "ground truth" in information systems to preserve the integrity of reputation mechanisms. Yet, the reliance on AI for decision-making risks turning humans into mere tools of automated systems. This reversal of agency calls for ethical frameworks that prioritize human dignity. Societal resilience depends on preserving human autonomy amidst technological upheaval.

7: Ethical and Philosophical Dimensions

Russell delves into ethical theories, particularly consequentialism, to guide AI's design and evaluation. Consequences for multiple stakeholders must be weighed equitably, reflecting utilitarian principles. However, he critiques narrow interpretations of utility that overlook moral complexity and societal dynamics. Positional goods, driven by pride and envy, exemplify zero-sum pursuits that undermine collective well-being. Russell warns against embedding such values into AI systems, advocating instead for broader considerations of human flourishing. Negative altruism, stemming from envy or malice, presents additional hurdles in aligning AI with societal goals. Philosophical debates on rationality, preference formation, and value alignment inform these discussions. AI must accommodate diverse human experiences without dictating them, preserving individual freedom. By integrating ethical insights, AI can become a tool for enhancing rather than constraining humanity. This requires ongoing dialogue between technologists, ethicists, and the public.

8: Governance and Regulation

Governance structures are pivotal in addressing AI's societal and ethical challenges. Russell critiques the fragmented landscape of advisory bodies and international panels, highlighting the need for actionable recommendations. Effective regulation must balance innovation with safety, avoiding reactionary policies that stifle progress. Industry resistance to oversight, driven by commercial interests, poses significant obstacles. Russell envisions a cultural shift that prioritizes collective well-being over individual gain, fostering sustainable innovation. He emphasizes proactive engagement with emerging technologies to mitigate risks before crises arise. Transparent standards and collaborative frameworks can build public trust in AI systems. However, governance efforts must also address global inequalities and power imbalances in AI development. A unified approach to regulation can ensure equitable access to AI's benefits. By embedding ethical principles into

governance, societies can harness AI's potential responsibly.

9: Human and Machine Collaboration

The integration of AI into human decision-making necessitates a rethinking of collaboration. Russell explores hierarchical planning as a means of simplifying complex tasks, mirroring human cognitive structures. Current methods, however, rely heavily on human input, limiting scalability. Advances in Bayesian reasoning and probabilistic models offer promise but remain constrained by representational challenges. Russell underscores the importance of building machines that learn from and adapt to human behavior dynamically. This requires bridging the gap between theoretical models and real-world complexities. Collaboration between humans and machines should enhance rather than replace human agency. Ethical safeguards must ensure that AI systems respect individual autonomy while supporting collective goals. By fostering mutual understanding, humans and machines can address global challenges effectively. Russell envisions a future where AI complements human ingenuity rather than supplanting it.

10: Toward a Beneficial AI Future

Russell concludes with a vision for a future shaped by beneficial AI. This entails designing systems that prioritize human preferences and values, learning adaptively from interactions. The principles of deference, uncertainty, and collaboration guide this approach. Russell cautions against simplistic notions of control, advocating for nuanced frameworks that accommodate uncertainty. Achieving these goals requires interdisciplinary efforts, uniting technologists, ethicists, and policymakers. Education and cultural shifts play a crucial role in preparing societies for AI's transformative impact. By embracing a long-term perspective, humanity can navigate the risks and opportunities of superintelligent AI. Russell remains optimistic about our ability to align AI with human interests through careful planning. His work underscores the importance of foresight, collaboration, and ethical responsibility in shaping AI's role in society. Together, these principles lay the foundation for a harmonious coexistence between humans and machines.

Excerpts

Stelle: 358

“Success [of a superintelligent AI] would be the biggest event in human history ... and perhaps the last event in human history.”

Stelle: 429

Betting against human ingenuity is foolhardy, particularly when our future is at stake. Within the AI community, a kind of denialism is emerging, even going as far as denying the possibility of success in achieving the long-term goals of AI.

Stelle: 439

A more predictable user can be fed items that they are likely to click on, thereby generating more revenue. People with more extreme political views tend to be more predictable in which items they will click on. (Possibly there is a category of articles that die-hard centrists are likely to click on, but it's not easy to imagine what this category consists of.) Like any rational entity, the algorithm learns how to modify the state of its environment—in this case, the user's mind—in order to maximize its own reward.⁸ The consequences include the resurgence of fascism, the dissolution of the social contract that underpins democracies around the world, and potentially the end of the European Union and NATO.

Stelle: 449

The concept of intelligence is central to who we are—that's why we call ourselves *Homo sapiens*, or “wise man.” After more than two thousand years of self-examination, we have arrived at a characterization of intelligence that can be boiled down to this: Humans are intelligent to the extent that our actions can be expected to achieve our objectives. All those other characteristics of intelligence—perceiving, thinking, learning, inventing, and so on—can be understood through their contributions to our ability to act successfully.

Stelle: 477

For thousands of years, we have known the perils of getting exactly what you wish for.

Stelle: 491

Uncertainty about objectives implies that machines will necessarily defer to humans: they will ask permission, they will accept correction, and they will allow themselves to be switched off. Removing the assumption that machines should have a definite objective means that we will need to tear out and replace part of the foundations of artificial intelligence—the basic definitions of what we are trying to do.

Stelle: 551

No one in AI is working on making machines conscious, nor would anyone know where to start, and no behavior has consciousness as a prerequisite.

Stelle: 565

One reason we understand the brain's reward system is that it resembles the method of reinforcement learning developed in AI, for which we have a very solid theory.

Stelle: 568

Organisms that are more effective in seeking reward—that is, finding delicious food, avoiding pain, engaging in sexual activity, and so on—are more likely to propagate their genes. It is extraordinarily difficult for an organism to decide what actions are most likely, in the long run, to result in successful propagation of its genes, so evolution has made it easier for us by providing built-in signposts.

Stelle: 584

The connection between learning and evolution was proposed in 1896 by the American psychologist James Baldwin

Stelle: 592

Provided the adaptive organisms can survive while learning, it seems that the capability for learning constitutes an evolutionary shortcut. Computational simulations suggest that the Baldwin effect is real.

Stelle: 596

The story of the Baldwin effect is fascinating but incomplete: it assumes that learning and evolution necessarily point in the same direction. That is, it assumes that whatever internal feedback signal defines the direction of learning within the organism is perfectly aligned with evolutionary fitness.

Stelle: 655

Utility exhibits diminishing returns with respect to money. This means that the utility of a given amount of money is not strictly proportional to the amount but grows more slowly. For example, the utility of having \$1,000,000,100 is much less than a hundred times the utility of having \$10,000,000. How much less? You can ask yourself! What would the odds of winning a billion dollars have to be for you to give up a guaranteed ten million? I asked this question of the graduate students in my class and their answer was around 50 percent, meaning that bet B would have an expected value of \$500 million to match the desirability of bet A. Let me say that again: bet B would have an expected dollar value fifty times greater than bet A, but the two bets would have equal utility.

Stelle: 665

John von Neumann (a great mathematician after whom the standard “von Neumann architecture” for computers was named) and Oskar Morgenstern published an axiomatic basis for utility theory. What this means is the following: as long as the preferences exhibited by an individual satisfy certain basic axioms that any rational agent should satisfy, then necessarily the choices made by that individual can be described as maximizing the expected value of a utility function. In short, a rational agent acts so as to maximize expected utility.

Stelle: 700

Maximizing expected utility may not require calculating any expectations or any utilities. It's a purely external description of a rational entity.

Stelle: 708

The species benefits from the presence of some fraction of individuals who are willing to sacrifice themselves in battle, or to go off on wild, speculative voyages of exploration, or to nurture the offspring of others. In such cases, an analysis of rationality that focuses entirely on the individual is clearly missing something essential.

Stelle: 716

A great deal of our cognitive structure is there to compensate for the mismatch between our small, slow brains and the incomprehensibly huge complexity of the decision problem that we face all the time.

Stelle: 718

So, while it would be quite unreasonable to base a theory of beneficial AI on an assumption that humans are rational, it's quite reasonable to suppose that an adult human has roughly consistent preferences over future lives. That is, if you were somehow able to watch two movies, each describing in sufficient detail and breadth a future life you might lead, such that each constitutes a virtual experience, you could say which you prefer, or express indifference.²² This claim is perhaps stronger than necessary, if our only goal is to make sure that sufficiently intelligent machines are not catastrophic for the human race.

Stelle: 900

The halting problem: Can an algorithm decide if a given program has an “infinite loop” that prevents it from ever finishing? Turing's proof that no algorithm can solve the halting problem is incredibly important for the foundations of mathematics, but it seems to have no bearing on the issue of whether computers can be intelligent.

Stelle: 907

Focusing on decidable problems, then, seems not to place any real restrictions on AI.

Stelle: 918

Exponential growth in difficulty means that problems may be solvable in theory (that is, they are certainly decidable) but sometimes unsolvable in practice; we call such problems intractable.

Stelle: 926

Complexity means that the real-world decision problem—the problem of deciding what to do right now, at every instant in one's life—is so difficult that neither humans nor computers will ever come close to finding perfect solutions.

Stelle: 928

This has two consequences: first, we expect that, most of the time, real-world decisions will be at best halfway decent and certainly far from optimal; second, we expect that a great deal of the mental architecture of humans and computers—the way their decision processes actually operate—will be designed to overcome complexity to the extent possible—that is, to make it possible to find even halfway decent answers despite the overwhelming complexity of the world.

Stelle: 1,064

AlphaZero has hard limitations: it works only in the class of discrete, observable, two-player games with known rules. The approach simply won't work at all for driving, teaching, running a government, or taking over the world.

Stelle: 1,119

In 1930, the brilliant Austrian logician Kurt Gödel had published his famous completeness theorem,⁵³ proving that there is an algorithm with the following property:⁵⁴ For any collection of knowledge and any question expressible in first-order logic, the algorithm will tell us the answer to the question if there is one.

Stelle: 1,204

Games such as Go and Dota 2 are a good testing ground for reinforcement learning methods because the reward function comes with the rules of the game. The real world is less convenient, however, and there have been dozens of cases in which faulty definitions of rewards led to weird and unanticipated behaviors. Some are innocuous, like the simulated evolution system that was supposed to evolve fast-moving creatures but in fact produced creatures that were enormously tall and moved fast by falling over. Others are less innocuous, like the social-media click-through optimizers that seem to be making a fine mess of our world.

Stelle: 1,422

learning algorithms can operate on encrypted data using the techniques of secure multiparty computation, so that users can benefit from pooling without compromising privacy in any way. Will software providers adopt privacy-preserving technology voluntarily, without legislative encouragement? That remains to be seen.

1,637

Deep learning systems are mostly data driven: at best, we can “wire in” some very weak forms of prior knowledge in the structure of the network. Probabilistic programming systems do allow for prior knowledge in the learning process, as expressed in the structure and vocabulary of the probabilistic knowledge base, but we do not yet have effective methods for generating new concepts and relationships and using them to expand such a knowledge base.

Stelle: 1,919

If human happiness requires being in the top 1 percent, then 99 percent of humans are going to be unhappy, even when the bottom 1 percent has an objectively splendid lifestyle. It will be important, then, for our cultures to gradually down-weight pride and envy as central elements of perceived self-worth.

Stelle: 1,941

Increasingly, AI systems are able to understand the content of what people are saying and doing, whether in speech, text, or video surveillance. In regimes where this technology is adopted for the purposes of control, it will be as if every citizen had their own personal Stasi operative watching over them twenty-four hours a day.

Stelle: 1,959

A more subtle way to change people's behavior is to modify their information environment so that they believe different things and make different decisions. Of course, advertisers have been doing this for centuries as a way of modifying the purchasing behavior of individuals. Propaganda as a tool of war and political domination has an even longer history. So what's different now? First, because AI systems can track an individual's online reading habits, preferences, and likely state of knowledge, they can tailor specific messages to maximize impact on that individual while minimizing the risk that the information will be disbelieved. Second, the AI system knows whether the individual reads the message, how long they spend reading it, and whether they follow additional links within the message. It then uses these signals as immediate feedback on the success or failure of its attempt to influence each individual; in this way, it quickly learns to become more effective in its work. This is how content selection algorithms on social media have had their insidious effect on political opinions.

Stelle: 1,972

AI systems can generate millions of false identities—the so-called bot armies—that can pump out billions of comments, tweets, and recommendations daily, swamping the efforts of mere humans to exchange truthful information.

Stelle: 1,977

The temptation for a government, particularly one with a top-down, engineering mind-set, is to reason as follows: it would be better if everyone behaved well, had a patriotic attitude, and contributed to the progress of the country; technology enables measurement of individual behavior, attitudes, and contributions; therefore, everyone will be better off if we set up a technology-based system of monitoring and control based on rewards and punishments.

Stelle: 1,982

Every act of kindness ceases to be an act of kindness and becomes instead an act of personal score maximization and is perceived as such by the recipient.

Stelle: 1,994

Everyone should also have the right to mental security—the right to live in a largely true information environment.

Stelle: 2,004

Rather than pursuing the idea that there is no freedom of thought without access to true information, democracies seem to have placed a naïve trust in the idea that the truth will win out in the end, and this trust has left us unprotected. Germany is an exception; it recently passed the Network Enforcement Act, which requires content platforms to remove proscribed hate speech and fake news, but this has come under considerable criticism as being unworkable and undemocratic.

Stelle: 2,017

One way to protect the functioning of reputation systems is to inject sources that are as close as possible to ground truth. A single fact that is certainly true can invalidate any number of sources that are only somewhat trustworthy, if those sources disseminate information contrary to the known fact. In many countries, notaries function as sources of ground truth to maintain the integrity of legal and real-estate information; they are usually disinterested third parties in any transaction and are licensed by governments or professional societies.

Stelle: 2,037

Information providers—at least those for who reputation matters—have an incentive to join industry associations and subscribe to codes of conduct that favor truth telling. In turn, social media platforms can offer users the option of seeing content from only reputable sources that subscribe to these codes and subject themselves to third-party fact-checking.

Stelle: 2,075

More convincing reason to reject autonomous weapons is that they are scalable weapons of mass destruction.

Stelle: 2,190

We need a plausible destination in order to plan a transition—that is, we need a plausible picture of a desirable future economy where most of what we currently call work is done by machines.

Stelle: 2,217

The connection between striving and enjoying is a central theme for our understanding of how to fashion a desirable future.

Stelle: 2,221

Inevitably, most people will be engaged in supplying interpersonal services that can be provided—or which we prefer to be provided—only by humans. That is, if we can no longer supply routine physical labor and routine mental labor, we can still supply our humanity.

Stelle: 2,227

Keynes: It will be those peoples, who can keep alive, and cultivate into a fuller perfection, the art of life itself and do not sell themselves for the means of life, who will be able to enjoy the abundance when it comes.

Stelle: 2,305

In the EU, Article 22 of the 2018 General Data Protection Regulation, or GDPR, explicitly forbids the granting of authority to machines in such cases: The data subject shall have the right not to be subject to a decision based solely on automated processing, including profiling, which produces legal effects concerning him or her or similarly significantly affects him or her. Although this sounds admirable in principle, it remains to be seen—at least at the time of writing—how much impact this will have in practice. It is often so much easier, faster, and cheaper to leave the decisions to the machine.

Stelle: 2,348

The question is whether the computer system remains a tool of humans, or humans become tools of the computer system—supplying information and fixing bugs when necessary, but no longer understanding in any depth how the whole thing is

Stelle: 2,440

In 1950, Wiener published *The Human Use of Human Beings*,⁵ whose front-cover blurb reads, “The ‘mechanical brain’ and similar machines can destroy human values or enable us to realize them as never before.”⁶ He gradually refined his ideas over time and by 1960 had identified one core issue: the impossibility of defining true human purposes correctly and completely. This, in turn, means that what I have called the standard model—whereby humans attempt to imbue machines with their own purposes—is destined to fail.

Stelle: 2,449

A technical way of saying this is that we may suffer from a failure of value alignment—we may, perhaps inadvertently, imbue machines with objectives that are imperfectly aligned with our own.

Stelle: 2,478

The reinforcement learning algorithms that optimize social-media click-through have no capacity to reason about human behavior—in fact, they do not even know in any meaningful sense that humans exist. For machines with much greater understanding of human psychology, beliefs, and motivations, it should be relatively easy to gradually guide us in directions that increase the degree of satisfaction of the machine’s objectives. For example, it might reduce our energy consumption by persuading us to have fewer children, eventually—and inadvertently—achieving the dreams of anti-natalist philosophers who wish to eliminate the noxious impact of humanity on the natural world.

Stelle: 2,495

Unfortunately, with superintelligent systems that can have a global impact, there are no simulators and no do-overs. It’s certainly very hard, and perhaps impossible, for mere humans to anticipate and rule out in advance all the disastrous ways the machine could choose to achieve a specified objective. Generally speaking, if you have one goal and a superintelligent machine has a different, conflicting goal, the machine gets what it wants and you don’t.

Stelle: 2,565

Value, for humans, is defined primarily by conscious human experience. If there are no humans and no other conscious entities whose subjective experience matters to us, there is nothing of value occurring.

Stelle: 2,797

To varying degrees, all the major technological issues of the twentieth century—nuclear power, genetically modified organisms (GMOs), and fossil fuels—succumbed to tribalism. On each issue, there are two sides, pro and anti. The dynamics and outcomes of each have been different, but the symptoms of tribalism are similar: mutual distrust and denigration, irrational arguments, and a refusal to concede any (reasonable) point that might favor the other tribe.

Stelle: 2,922

Intelligence and final goals are orthogonal: more or less any level of intelligence could in principle be combined with more or less any final goal.

Stelle: 2,965

The “skeptic” position seems to be that, although we should probably get a couple of bright people to start working on preliminary aspects of the problem, we shouldn’t panic or start trying to ban AI research. The “believers,” meanwhile, insist that although we shouldn’t panic or start trying to ban AI research, we should probably get a couple of bright people to start working on preliminary aspects of the problem.

Stelle: 2,973

The debate has highlighted the conundrum we face: if we build machines to optimize objectives, the objectives we put into the machines have to match what we want, but we don’t know how to define human objectives completely and correctly. Fortunately, there is a middle way.

Stelle: 2,979

Let’s remind ourselves of the task at hand: to design machines with a high degree of intelligence—so that they can help us with difficult problems—while ensuring that those machines never behave in ways that make us seriously unhappy. The task is, fortunately, not the following: given a machine that possesses a high degree of intelligence, work out how to control it. If that were the task, we would be toast.

Stelle: 2,989

Like many other fields, AI has adopted the standard model: we build optimizing machines, we feed objectives into them, and off they go. That worked well when the machines were stupid and had a limited scope of action; if you put in the wrong objective, you had a good chance of being able to switch off the machine, fix the problem, and try again. As machines designed according to the standard model become more intelligent, however, and as their scope of action becomes more global, the approach becomes untenable. Such machines will pursue their objective, no matter how wrong it is; they will resist attempts to switch them off; and they will acquire any and all resources that contribute to achieving the objective.

Stelle: 3,000

Principles for Beneficial Machines

The machine’s only objective is to maximize the realization of human preferences. The machine is initially uncertain about what those preferences are. The ultimate source of information about human preferences is human behavior.

Stelle: 3,056

In all the work on utility maximization, goal achievement, cost minimization, reward maximization, and loss minimization, it is assumed that the utility function, the goal, the cost function, the reward function, and the loss function are known perfectly.

Stelle: 3,074

Human choices reveal information about human preferences.

Stelle: 3,077

Humans are not perfectly rational: imperfection comes between human preferences and human choices, and the machine must take into account those imperfections if it is to interpret human choices as evidence of human preferences.

Stelle: 3,085

Unfortunate conflict between the commonsense meaning of value and the more technical sense in which it is used in economics, AI, and operations research. In ordinary usage, values are what one uses to help resolve moral dilemmas; as a technical term, on the other hand, value is roughly synonymous with utility,

Stelle: 3,089

To avoid this confusion, the principles talk about human preferences rather than human values, since the former term seems to steer clear of judgmental preconceptions about morality.

Stelle: 3,091

“Putting in values” is, of course, exactly the mistake I am saying we should avoid, because getting the values (or preferences) exactly right is so difficult and getting them wrong is potentially catastrophic. I am proposing instead that machines learn to predict better, for each person, which life that person would prefer, all the while being aware that the predictions are highly uncertain and incomplete.

Stelle: 3,155

There is a lesson in Asilomar for all of science: the best way to respond to concerns created by emerging knowledge or early-stage technologies is for scientists from publicly funded institutions to find common cause with the wider public about the best way to regulate—as early as possible. Once scientists from corporations begin to dominate the research enterprise, it will simply be too late.

Stelle: 3,183

We will want, eventually, to prove theorems to the effect that a particular way of designing AI systems ensures that they will be beneficial to humans.

Stelle: 3,253

When a proof of safety succeeds, we need to make sure it’s not succeeding because we have made unrealistically strong assumptions or because the definition of safety is too weak. When a proof of safety fails, we need to resist the temptation to strengthen the assumptions to make the proof go through—for example, by adding the assumption that the program’s code remains fixed. Instead, we need to tighten up the design of the AI system—for example, by ensuring that it has no incentive to modify critical parts of its own code.

Stelle: 3,296

Perhaps the easiest way to understand IRL (Inverse Reinforcement Learning) is this: the observer starts with some vague estimate of the true reward function and then refines this estimate, making it more precise, as more behavior is observed. Or, in Bayesian language: start with a prior probability over possible reward functions and then update the probability distribution on reward functions as evidence arrives.

Stelle: 3,324

With a human and a robot in the same environment, we are in the realm of game theory—just as in the penalty shoot-out between Alice and Bob on page 28. We assume, in this first version of the theory, that the human has preferences and acts according to those preferences. The robot doesn't know what preferences the human has, but it wants to satisfy them anyway. We'll call any such situation an assistance game, because the robot is, by definition, supposed to be helpful to the human.

Stelle: 3,446

Always allocate some probability, however small, to preferences that are logically possible.

Stelle: 3,475

Loophole principle: if a sufficiently intelligent machine has an incentive to bring about some condition, then it is generally going to be impossible for mere humans to write prohibitions on its actions to prevent it from doing so or to prevent it from doing something effectively equivalent.

Stelle: 3,477

The best solution for preventing tax avoidance is to make sure that the entity in question wants to pay taxes. In the case of a potentially misbehaving AI system, the best solution is to make sure it wants to defer to humans.

Stelle: 3,550

If we propose that an AI system can be trained to behave well through reinforcement learning, with humans giving feedback signals that define the direction of improvement, the inevitable result is that the AI system works out how to control the humans and forces them to give maximal positive rewards at all times. You might think that this would just be a form of pointless self-delusion on the part of the AI system, and you'd be right. But it's a logical consequence of the way reinforcement learning is defined. The process works fine when the reward signal comes from "outside the universe" and is generated by some process that can never be modified by the AI system; but it fails if the reward-generating process (that is, the human) and the AI system inhabit the same universe.

Stelle: 3,593

Discussions beginning with "suppose that robot R has purpose P" are fine for gaining some intuition about how things might unfold, but they cannot lead to theorems about real machines. We need much more nuanced and precise definitions of purposes in machines before we can obtain guarantees of how they will behave over the long term.

Stelle: 3,652

Satisfying the preferences of an individual has implications for other people, depending on the individual's preferences about the well-being of others.

Stelle: 3,683

Societies will find themselves passing hundreds of new laws every day to counteract all the loopholes that machines will find in existing laws. Humans tend not to take advantage of these loopholes, either because they have a general understanding of the underlying moral principles or because they lack the ingenuity required to find the loopholes in the first place.

Stelle: 3,692

The approach that is most relevant for understanding how AI systems should be designed is often called consequentialism: the idea that choices should be judged according to expected consequences. The two other principal approaches are deontological ethics and virtue ethics, which are, very roughly, concerned with the moral character of actions and individuals, respectively, quite apart from the consequences of choices.³ Absent any evidence of self-awareness on the part of machines, I think it makes little sense to build machines that are virtuous or that choose actions in accordance with moral rules if the consequences are highly undesirable for humanity. Put another way, we build machines to bring about consequences, and we should prefer to build machines that bring about consequences that we prefer.

Stelle: 3,720

We plan to build consequentialist machines, the next question is how to evaluate consequences that affect multiple people. One plausible answer is to give equal weight to everyone's preferences—in other words, to maximize the sum of everyone's utilities. This answer is usually attributed to the eighteenth-century British philosopher Jeremy Bentham and his pupil John Stuart Mill, who developed the philosophical approach of utilitarianism.

Stelle: 3,738

The economist John Harsanyi propounded this view with his principle of preference autonomy: In deciding what is good and what is bad for a given individual, the ultimate criterion can only be his own wants and his own preferences. Harsanyi's preference utilitarianism is therefore roughly consistent with the first principle of beneficial AI, which says that a machine's only purpose is the realization of human preferences. AI researchers should definitely not be in the business of deciding what human preferences should be!

Stelle: 3,779

The AI community should pay careful attention to the thrusts and counterthrusts of philosophical and economic debates on utilitarianism because they are directly relevant to the task at hand. Two of the most important, from the point of view of designing AI systems that will benefit multiple individuals, concern interpersonal comparisons of utilities and comparisons of utilities across different population sizes. Both of these debates have been raging for 150 years or more, which leads one to suspect their satisfactory resolution may not be entirely straightforward.

Stelle: 3,834

Machines with sufficient foresight may be able to consider courses of action leading to different population sizes, just as the Chinese government did with its one-child policy in 1979. It's quite likely, for example, that we will be asking AI systems for help in devising solutions for global climate change—and those solutions may well involve policies that tend to limit or even reduce population size.

Stelle: 3,907

In his paper that introduced preference utilitarianism, Harsanyi attributes negative altruism to "sadism, envy, resentment, and malice" and argues that they should be ignored in calculating the sum total of human utility in a population: No amount of goodwill to individual X can impose the moral obligation on me to help him in hurting a third person, individual Y. This seems to be one area in which it is reasonable for the designers of intelligent machines to put a (cautious) thumb on the scales of justice, so to speak.

Stelle: 3,912

Unfortunately, negative altruism is far more common than one might expect. It arises not so much from sadism and malice³⁰ but from envy and resentment and their converse emotion, which I will call pride (for want of a better word).

Stelle: 3,927

1977, the British economist Fred Hirsch published *The Social Limits to Growth*,³³ in which he introduced the idea of positional goods. A positional good is anything—it could be a car, a house, an Olympic medal, an education, an income, or an accent—that derives its perceived value not just from its intrinsic benefits but also from its relative properties, including the properties of scarcity and being superior to someone else's. The pursuit of positional goods, driven by pride and envy, has the character of a zero-sum game,

Stelle: 3,974

One obvious difference between humans and rational entities is that, at any given moment, we are not choosing among all possible first steps of all possible future lives. Not even close. Instead, we are typically embedded in a deeply nested hierarchy of “subroutines.” Generally speaking, we are pursuing near-term goals rather than maximizing preferences over future lives, and we can act only according to the constraints of the subroutine we're in at present.

Stelle: 3,983

More generally, learning about human preferences seems to require learning about the actual structure of human lives. What are all the things that we humans can be engaged in, either singly or jointly? What activities are characteristic of different cultures and types of individuals?

Stelle: 4,024

It's a bad idea to identify human preferences with simple choices between incompletely described options that are intractable to evaluate and include elements of unknown desirability. Such choices provide indirect evidence of underlying preferences, but they are not constitutive of those preferences. That's why I have couched the notion of preferences in terms of future lives—for example by imagining that you could experience, in a compressed form, two different movies of your future life and then express a preference between them (see page 26). The thought experiment is of course impossible to carry out in practice, but one can imagine that in many cases a clear preference would emerge long before all the details of each movie had been filled in and fully experienced. You may not know in advance which you will prefer, even given a plot summary; but there is an answer to the actual question, based on who you are now, just as there is an answer to the question of whether you will like durian when you try it.

Stelle: 4,047

Some psychologists have called into question the very notion that there is one self whose preferences are sovereign in the way that Harsanyi's principle of preference autonomy suggests. Most prominent among these psychologists is my former Berkeley colleague Daniel Kahneman.

Stelle: 4,098

The possibility of changes in the typical preferences of human populations over historical time naturally focuses attention on the question of how each individual's preferences are formed and the plasticity of adult preferences.

Stelle: 4,105

Preference change presents a challenge for theories of rationality at both the individual and societal level. For example, Harsanyi's principle of preference autonomy seems to say that everyone is entitled to whatever preferences they have and no one else should touch them. Far from being untouchable, however, preferences are touched and modified all the time, by every experience a person has. Machines cannot help but modify human preferences, because machines modify human experiences.

Stelle: 4,168

Aristotle himself wrote, "The main concern of politics is to engender a certain character in the citizens and to make them good and disposed to perform noble actions." Let's just say that there are risks associated with intentional preference engineering on a global scale. We should proceed with extreme caution.

Stelle: 4,219

If the sheer number of initiatives to develop effective governance of AI is any guide, then we are in excellent shape. Everyone and their uncle is setting up a Board or a Council or an International Panel. The World Economic Forum has identified nearly three hundred separate efforts to develop ethical principles for AI. My email inbox can be summarized as one long invitation to the Global World Summit Conference Forum on the Future of International Governance of the Social and Ethical Impacts of Emerging Artificial Intelligence Technologies.

Stelle: 4,241

The G7 has proposed an International Panel on Artificial Intelligence, hoping that it will grow into something like the UN's Intergovernmental Panel on Climate Change.

Stelle: 4,244

Many governments around the world are equipping themselves with advisory bodies to help with the process of developing regulations; perhaps the most prominent example is the EU's High-Level Expert Group on Artificial Intelligence.

Stelle: 4,251

At present, there are no implementable recommendations that can be made to governments or other organizations considering the issue of maintaining control over AI systems. A regulation such as "AI systems must be safe and controllable" would carry no weight, because these terms do not yet have precise meanings and because there is no widely known engineering methodology for ensuring safety and controllability.

Stelle: 4,270

Inevitably, however, the tech industry is going to have to acknowledge that its products matter; and, if they matter, then it matters that the products not have harmful effects. This means that there will be rules governing the nature of interactions with humans, prohibiting designs that, say, consistently manipulate preferences or produce addictive behavior. I have no doubt that the transition from an unregulated to a regulated world will be a painful one. Let's hope it doesn't require a Chernobyl-sized disaster (or worse) to overcome the industry's resistance.

Stelle: 4,281

Already, we are losing the battle against malware and cybercrime. (A recent report estimates over two billion victims and an annual cost of around \$600 billion.

Stelle: 4,314

There is a tragedy of the commons at work here: for any individual human, it may seem pointless to engage in years of arduous learning to acquire knowledge and skills that machines already have; but if everyone thinks that way, the human race will, collectively, lose its autonomy.

Stelle: 4,316

The solution to this problem seems to be cultural, not technical. We will need a cultural movement to reshape our ideals and preferences towards autonomy, agency, and ability and away from self-indulgence and dependency—if you like, a modern, cultural version of ancient Sparta's military ethos. This would mean human preference engineering on a global scale along with radical changes in how our society works.

Stelle: 4,469

In the real world, we would be lost without a vast library of actions at dozens of levels of abstraction. (In the game of Go, there is no obvious hierarchy of actions, so most of us are lost.) At present, however, all existing methods for hierarchical planning rely on a human-generated hierarchy of abstract and concrete actions; we do not yet understand how such hierarchies can be learned from experience.

Stelle: 4,630

Bayesian networks provide machinery for representing the necessary probabilities and performing the calculations to implement Bayesian updating for many complex tasks. Like propositional logic, however, they are quite limited in their ability to represent general knowledge. In many applications, the Bayesian network representation becomes very large and repetitive

Stelle: 4,646

The combination of first-order logic and probability actually gives us much more than just a way to express uncertain information about lots of objects. The reason is that when we add uncertainty to worlds containing objects, we get two new kinds of uncertainty: not just uncertainty about which facts are true or false but also uncertainty about what objects exist and uncertainty about which objects are which.

Stelle: 4,844

As yet, we have very little understanding as to why deep learning works as well as it does. Possibly the best explanation is that deep networks are deep: because they have many layers, each layer can learn a fairly simple transformation from its inputs to its outputs, while many such simple transformations add up to the complex transformation required to go from a photograph to a category label. In addition, deep networks for vision have built-in structure that enforces translation invariance and scale invariance—meaning that a dog is a dog no matter where it appears in the image and no matter how big it appears in the image.

Stelle: 4,964

The presence or absence of consciousness—actual subjective experience—certainly makes a difference in our moral consideration for machines. If ever we gain enough understanding to design conscious machines or to detect that we have done so, we would face many important moral issues for which we are largely unprepared.

Stelle: 5,146

A compendium of cases in the literature where misspecification of reward functions led to unexpected behavior: Victoria Krakovna, “Specification gaming examples in AI,” Deep Safety (blog), April 2, 2018.

Stelle: 5,328

Goodhart’s law: “Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes.”

Stelle: 5,684

For an excellent machine-oriented introduction to standard ethical theories and their implications for designing AI systems, see Wendell Wallach and Colin Allen, *Moral Machines: Teaching Robots Right from Wrong* (Oxford University Press, 2008).